

title

# Material Specification Sheet

text

**Name:** PEEK, PolyEtherEther-Ketone

text

**Supplier:** Plastics Supplier Company

title

## Material description

text

PEEK is a high-performance engineering thermoplastic. It is strong, rigid, and highly resistant to chemicals, water, steam, and elevated temperatures. Because of this combination of properties, PEEK is commonly used in demanding environments where ordinary plastics would not perform well. In some harsh-service applications, it may also be used as a strong alternative to fluoropolymers.

title

## Main benefits

text

PEEK provides strong chemical resistance, low moisture uptake, good wear and abrasion resistance, high mechanical strength, and reliable performance in hot water or steam. It also has a UL94 V-0 flammability rating.

title

## Common applications

text

Typical application areas include automotive, marine, electronics, medical, aerospace, oil and gas, and alternative energy.

title

## Key properties

table

| Property                      | Value                                                 |
|-------------------------------|-------------------------------------------------------|
| Density                       | ~1.31-1.32 g/cm <sup>3</sup>                          |
| Water absorption, 24 h        | 0.10-0.15%                                            |
| Tensile strength              | 14,500-16,000 psi, unfilled PEEK                      |
| Tensile modulus               | 450,000-550,000 psi, unfilled PEEK                    |
| Flexural strength             | 24,000-25,000 psi, unfilled PEEK                      |
| Compressive strength          | 17,000-20,000 psi, unfilled PEEK                      |
| Melting temperature           | 644°F / 340°C                                         |
| Maximum operating temperature | 480°F / 249°C, unfilled PEEK                          |
| Volume resistivity            | 4.9 x 10 <sup>16</sup> ohm-cm, extruded unfilled PEEK |